PITT | SWANSON ENGINEERING

Thera-Comp

Brody Fiorito, Suhani Sengupta, Shreya Sharma, Kanen Patel

WHAT IS COMPRESSION THERAPY?

Compression therapy applies controlled external pressure to a limb to promote venous return and reduce fluid.

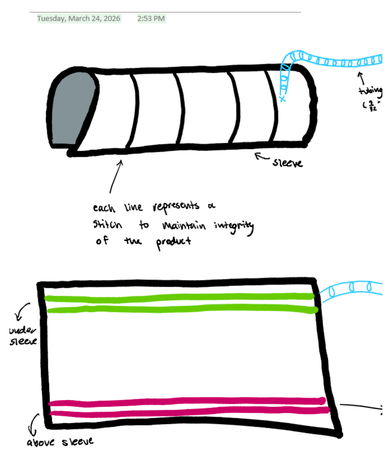
Commercial compression devices are effective but expensive and inaccessible in low-resource settings.

This project replicates that therapeutic function using only locally sourced materials, using no electronics and no custom components.

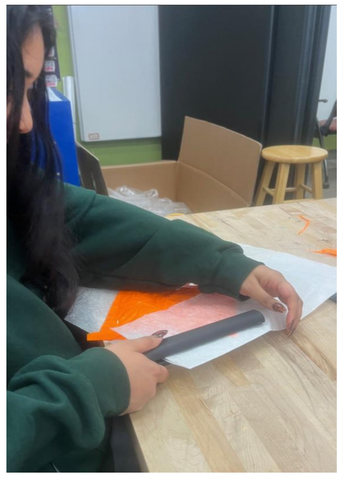
RESEARCH & BACKGROUND

- Inspired by NormaTec, a leading pneumatic compression system at a cost inaccessible for most users
- Clinical literature established a safe therapeutic pressure range of 15-40mmHg
- Electric inflation ruled out early: adds cost and undermines the low-resource design goal
- Interviewed occupational and physical therapists, a PT student, two NCAA athletes, and several competitive athletes to ground the design in real user needs

Final Product



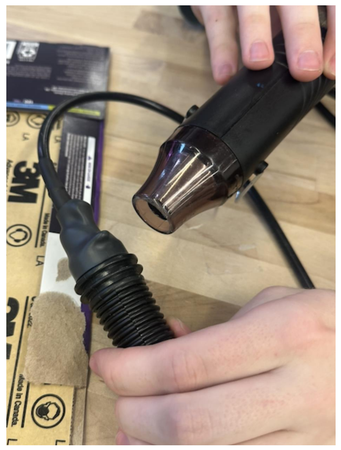
1. Sketch of prototype



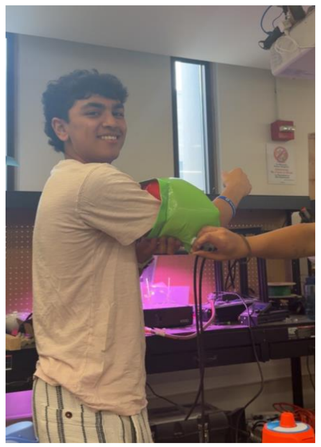
2. Heat sealing bladder

3. Silicone-sealing valves

4. Air Bladder



5. Heat sealing pump



6. Testing Air Bladder

MATERIALS & MANUFACTURING

- Sleeve has internal air bladder & external fabric shell
- Bladder made from a large beach ball cut and heat sealed using a hair straightener
- Contains 3 barbed fittings for inflation, pressure monitoring, and pressure release

SUSTAINABILITY

- All materials chosen to be low-cost and widely available, no specialized materials required
- Designed so that individuals in low-resource and low-income communities could realistically build it themselves
- Core components are everyday household or hardware store items

MANUFACTURING GUIDE